

TP1

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Part 1

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Part 1

The MLP has 784 inputs, hidden layers of 256 and 128 neurons, and 10 outputs, for 235,146 parameters. I used cross-entropy on the logits, then trained for 15 epochs with Adam ($lr=1e-3$, batch size 64, no weight decay).

The last epoch gave 94.15% train, 89.06% validation and 88.55% test accuracy. The baseline gap plot ends at 5.10 points. Validation loss rose from 0.317 at epoch 7 to 0.373 at epoch 15 while train loss fell, so it overfits.

Experiment 1

Hidden layers	Train	Validation	Gap	Test
[4]	81.94%	81.67%	0.28	80.79%
[64]	92.26%	88.28%	3.99	87.99%
[256, 128]	94.15%	89.06%	5.10	88.55%
[512, 256]	94.61%	89.41%	5.20	88.58%

Gaps are in percentage points. The [4] model underfits even with its small gap. The gap plot shows larger gaps as model size increases. In the validation accuracy plot, [512, 256] gains only 0.35 point over [256, 128]. I kept [256, 128] for the next tests because the larger model has more than twice as many parameters.

Experiment 2A

With SGD, $lr=1e-5$ gave 16.93% validation and 17.57% test accuracy. $1e-3$ gave 82.03% and 80.64%; 0.1 gave 89.40% and 88.89%. The learning rate plot shows almost no progress at $1e-5$. $1e-3$ learns much more slowly than 0.1 .

Experiment 2B

The best tested SGD rate was 0.1 (89.40% validation, epoch 15). For Adam it was $1e-3$ (89.38%, epoch 12). In the optimiser plot, Adam reached 88% at epoch 3, SGD at epoch 5. At $lr=1e-3$, Adam ended at 89.06% and SGD at 82.03%. Adam adapts its updates per parameter, but $lr=0.1$ still made it collapse to about 10%.

Experiment 3

Setting	Train	Validation	Gap	Test	Best epoch
None	95.18%	89.27%	5.91	88.55%	12
Dropout 0.3	90.61%	88.97%	1.64	88.58%	17
Weight decay $1e-3$	90.72%	88.73%	1.99	88.25%	16
Dropout with batch norm	91.51%	89.38%	2.12	89.31%	14

Gaps are in percentage points. Dropout switches off activations during training; weight decay limits large weights. In the regularisation gap plot, dropout has the smallest gap. Dropout with batch norm has the best final validation accuracy and peaks at 89.64%. Dropout and weight decay peak later than the run without regularisation. They fit the training set less, but validation accuracy changes little.